SEOYEONG JANG EECE Ph.D. Applicant

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RESEARCH INTERESTS

□ Ultra-low-power and energy-efficient mixed-signal ICs □ Reference-free and fully synthesizable sensors
□ PLL-based digital feedback □ Digital-assisted analog circuits □ Circuit design automation

EDUCATION

M.S. **KAIST,** Graduate School of AI Semiconductor — Daejeon, Korea
Sep. 2025 – *Advisor: Prof. Wanyeong Jung* | *GPA 3.84/4.3*

B.S. **Hanyang University,** Electrical Biomedical Engineering — Seoul, Korea
– Aug. 2025 *GPA 4.25/4.5, Summa Cum Laude*
Transferred from Duksung Women's University, Mar. 2023 (Biotechnology, Mar. 2021–) — Seoul, Korea

CURRENT RESEARCH — M.S. THESIS

Main research **Reference-Free Supply-Locked Ring-Oscillator Temperature Sensor**

- *±0.1 °C, −40 to 125 °C, XX µW, XX mm^2 (TBD)*
- Designed a custom standard-cell-based 2-stack LVT ring oscillator as the reference clock. A PLL-based feedback loop maintains a constant shared supply voltage for the LVT and HVT ring oscillators, and temperature is measured from their frequency ratio.
- Performed **end-to-end analog and digital IC design**, from transistor-level design to DRC/LVS-clean layout and system-level loop/noise analysis.

- **TSMC 28nm** tapeout in progress (GDS Sep. 2026); first-author paper in prep. (VLSI 2027)
  ***Synopsys flow**: Design Compiler, VCS, Verdi, IC Compiler II, PrimeTime, Calibre*
- **Samsung 14nm FinFET** tapeout selected (GDS Nov. 2026); first-author paper in prep. (ISSCC 2028)
  ***Cadence flow**: Genus, Innovus, Xcelium, SimVision, Tempus, Voltus, Pegasus*

Sub-project **Research Plugin and Skills for Circuit-Paper Analysis**

- Participating in the development of a lab-wide Codex research plugin for paper discovery, structured analysis, evidence extraction, and research-workspace management.
- Responsible for designing the RAG and retrieval component using keyword, semantic, schematic-topology, functional, and behavioral representations to retrieve related circuit techniques across different figures and terminology.

PUBLICATIONS

- Sunghwan Kim*, **Seoyeong Jang**, Rakesh Kumar Jha, Juwan Choi, "Naturally Derived Luminescent Material in Engineered Silk and Its Application as a Fluorescent Dye with a Large Stokes Shift and Sensing Capability," *ACS Biomaterials Science & Engineering*, vol. 10, no. 7, pp. 4552–4561, 2024. (IF 5.4, JIF rank 15/53) doi:10.1021/acsbiomaterials.4c00793

RESEARCH INTERNSHIPS

Jan.–Feb. 2025 **Kwon Research Group (Prof. Kyeong-Ha Kwon)** — Daejeon, Korea
Undergraduate Research, KAIST
- Designed a low-noise amplifier for electrochemical impedance spectroscopy (EIS) in a high-voltage battery-monitoring IC (BMIC) using systematic g_m/I_D sizing in Cadence Virtuoso. [summary]

Jun.–Aug. 2024 **Laboratory for Semiconductor System (Prof. Hoi-Jun Yoo)** — Daejeon, Korea
Undergraduate Research, PIM Semiconductor Design Research Center, KAIST
- Implemented real-time 3-D panoramic image projection for VR and optimized it for real-time computation on a mobile platform (Android/Java, PyTorch). [summary]

Jun. 2023– May 2024 **Laboratory for Advanced Biomaterials & Biodevices (Prof. Sung-Hwan Kim)** — Seoul, Korea
Undergraduate Research, Hanyang University
- Researched graphene/CNF E-tattoos for non-invasive drug delivery; won the Outstanding Research Award from the Dean of Biomedical Engineering. [summary]
- Researched biomedical applications and optical characteristics of silk-protein biomaterials; presented a poster at the 2023 Fall Meeting of the Korean Society of Medical & Biological Engineering; published in *ACS Biomaterials Science & Engineering* (IF 5.4). [summary]

SCHOLARSHIPS & HONORS

2025.11– **National STEM Master's Excellence Scholarship**, Korea Student Aid Foundation. ($3,500/year)
2025.08 **Graduation Excellence Award** from the University President (GPA 4.0+), Hanyang Univ.
2025.05 **Achievement Award** from the University President, 86th Anniversary Ceremony, Hanyang Univ.
2023–2025 **SeAH Haiam Academic Scholarship**, SeAH Haiam Scholarship Foundation. (Full tuition and stipend)
2023–2025 **Semester High Honors, 6 consecutive semesters**, Hanyang Univ. (2023.06–2025.12)
2020.06 **College Dean's Scholarship** for highest academic standing 1st Year(Ranked 1st), Duksung Women's Univ.

AWARDS

2025.06 3rd, **Excellence Award** (Univ. President), Running Pacemaker (Bio-Analog Circuit Tutor), Hanyang Univ.
2025.05 3rd, **Excellence Award**, ICT Convergence Project Competition, Device-mart (Korea).
2025.01 1st, **Grand Award** (Univ. President), Hanyang Academic Conference, Hanyang Univ.
2024.11 1st, **Minister of Education Award**, Microcontroller Application Competition, Seoul National Univ.
2024.10 **Best Question Award** (Univ. President), ASKTON (ASK+HACKATHON), Hanyang Univ.
2024.06 2nd, **Outstanding Research Award**, Capstone Design Poster Presentation, Hanyang Univ.
2021.02 1st, **Grand Award** (Univ. President), Computational Thinking Convergence Competition, Duksung Women's Univ.

PROJECT EXPERIENCE

Nov. 2024 **COSS Next-Generation Semiconductor MCU Application Competition — Team Leader**
Microcontroller Application Competition, Seoul National University (POLARIS)
– Built a bare-metal embedded 34-key digital piano; flashed firmware directly via USBasp. [summary]
– Stored LED control data in flash using PROGMEM, enabling updates for 486 NeoPixels per timer interval, **over 10× more than other teams**, while maintaining simultaneous animation and piano playback under tight MCU memory constraints.
– Won the **Minister of Education Award**; received the **Achievement Award** at Hanyang's Foundation Day Ceremony.

May–Nov. 2024 **Low-Cost Eye-Drawing Platform Using MCU-Based Electrooculography (EOG) — Team Leader**
Hanyang Academic Conference, Hanyang University
– Designed and built the analog front end for EOG measurement and a MATLAB algorithm mapping EOG signals to drawing movements.
– Won the **Grand Award** (Hanyang Academic Conference) and the Best Question Award (ASKTON) from the University President.

Apr.–Jun. 2024 **EMG-Based Prosthetic Arm Using MCU — Team Leader**
Term Project, Digital Engineering Course, Hanyang University
– Filtered and feature-extracted EMG/EOG signals on an ATmega128 and mapped them to prosthetic-limb movements.

TEACHING & OUTREACH

Mar. 2026– **President, Hanyang Prestigious Network (HPN)**
Alumni network for overseas Ph.D. admissions, Hanyang University
– Mentor **14 students** on SOPs, CVs, faculty outreach, and interviews; organizing the **university-wide information session on overseas graduate admissions** (May 2027).

Jan.–Dec. 2024 **Hanyang University Major Manager (HUMM) for High School Students** [summary]
Admissions Office, Hanyang University
– Delivered one-hour lectures on electronic-biomedical engineering research at **13 high schools** across Korea; certificate of completion from the University President. (30 hrs)

Apr.–Jun. 2024 **Tutor, Major-Course Tutoring for International Students** [summary]
College of Engineering, Hanyang University
– Taught a "Probability and Statistics" course **in English** to international students. (30 hrs)

Sep.–Nov. 2024 **Mentor, Cultural Exchange for International Students** [summary]
Institute of International Education, Hanyang University
– Taught Korean language and culture to international students. (30 hrs)

Jul.–Aug. 2024 **Volunteer Art Instructor for Children with Autism Spectrum Disorder**
Seoul Metropolitan Government–Hanyang University Art Education Support Program
– Supported students in art activities and encouraged participation. (30 hrs)